Sebastian Banuelos- Ae103 Final Project

SkyGuard Technologies

ADCS Preliminary Design Review

1 Introduction

At SkyGuard technologies we develop cutting edge tools for space exploration. More specifically, we engineer critical components for CubeSat satellites. CubeSat have demonstrated to be a revolutionary platform for affordable payloads in space. Ultimately, we are committed to further the capabilities of this vehicle.

SkyGuard's newest line of products is an Attitude Determination and Control System for Guidance Navigation and Control of 3U-6U CubeSats. Our system features 6DOF gyroscope and magnetometer to provide information regarding the satellite's attitude, and a star sensor to provide information regarding the satellite's orbital location. It also incorporates 3 control reaction wheels to correct the pointing attitude of the spacecraft, adapting to every customer's needs. The following document details the technical details behind our newest product, including example simulation of our control system at work.

2 Attitude Dynamics

In the following section we derive the equations of motion describing a CubeSat attitude dynamics.

From Euler's equation of motion we have

$$I\dot{\omega}_B^I + \omega_B^I \times (I\omega_B^I) = T \tag{1}$$

where ω_B^I is the angular velocity of the vehicle with respect to the inertial frame, I is the vehicle inertia, and T is the applied torques on the vehicle.

During the following sections, we express the vehicle's attitude in with respect to the orbital frame. Thus, we have

$$\omega_B^I = \omega_B^{OR} + \omega_{OR}^I \tag{2}$$

where ω_B^{OR} is the angular velocity of the vehicle with respect to the orbit and ω_{OR}^I is the angular velocity of the orbit with respect to the inertial frame.

Substituting (2) into (1)

$$I(\dot{\omega}_B^{OR} + \dot{\omega}_{OR}^I) + (\omega_B^{OR} + \omega_{OR}^I) \times I(\omega_B^{OR} + \omega_{OR}^I) = T$$

Assuming the satellite follows a circular orbit $\dot{\omega}_{OR}^I = 0$

$$I\dot{\omega}_B^{OR} + (\omega_B^{OR} + \omega_{OR}^I) \times I(\omega_B^{OR} + \omega_{OR}^I) = T$$

Now we transform the orbital angular velocity from the inertial frame to the body frame in quaternion representation

$$\begin{aligned}\left[\omega_{OR}^I\right]^B &= [C]_{OR}^B \left[\omega_{OR}^I\right]^{OR} \\ &= \omega_0 \begin{bmatrix} 1 - 2(q_2^2 + q_3^2) \\ 2(q_1 q_2 - q_3 q_4) \\ 2(q_1 q_3 + q_2 q_4) \end{bmatrix}\end{aligned}$$

where $\omega_0 = \sqrt{\mu/R_0^3}$ is the orbital frequency.

Thus, we have

$$I\dot{\omega}_B^{OR} + (\omega_B^{OR} + [C]_{OR}^B \left[\omega_{OR}^I\right]^{OR}) \times I(\omega_B^{OR} + [C]_{OR}^B \left[\omega_{OR}^I\right]^{OR}) = T$$

Introduce the following notation $\omega = \omega_B^{OR}$ and $\alpha = [C]_{OR}^B \left[\omega_{OR}^I\right]^{OR}$

$$I\dot{\omega} + (\omega + \alpha) \times I(\omega + \alpha) = T$$

Finally, we will consider two external torques applied on the spacecraft, the control torque T_c exerted by the reaction wheels, and the gravity gradient torque T_g caused by Earth's gravitational field acting on a non-symmetric spacecraft.

$$I\dot{\omega} + (\omega + \alpha) \times I(\omega + \alpha) = T_c + T_g \tag{3}$$

Note T_g is given in quaternion representation by

$$T_g = 3\omega_0^2 \begin{bmatrix} (I_z - I_y)2(q_2q_3 + q_1q_4)(1 - 2(q1^2 + q_2^2)) \\ (I_x - I_z)(1 - 2(q_1^2 + q_2^2))2(q_1q_3 - q_2q_4) \\ (I_y - I_x)2(q_1q_3 - q_2q_4)2(q_2q_3 + q_1q_4) \end{bmatrix}$$

Decompose equation (3) above into three principal orthogonal components.

$$I_x\dot{\omega}_x + I_z(\omega_y + \alpha_y)(\omega_z + \alpha_z) - I_y(\omega_z + \alpha_z)(\omega_y + \alpha_y) = T_x$$

$$I_y\dot{\omega}_y + I_x(\omega_z + \alpha_z)(\omega_x + \alpha_x) - I_z(\omega_x + \alpha_x)(\omega_z + \alpha_z) = T_y$$

$$I_z\dot{\omega}_z + I_y(\omega_x + \alpha_x)(\omega_y + \alpha_y) - I_x(\omega_y + \alpha_y)(\omega_x + \alpha_x) = T_z$$

Rearrange

$$I_x\dot{\omega}_x + (\omega_y + \alpha_y)(\omega_z + \alpha_z)(I_z - I_y) = T_{c,x} + T_{g,x}$$

$$I_y\dot{\omega}_y + (\omega_z + \alpha_z)(\omega_x + \alpha_x)(I_x - I_z) = T_{c,y} + T_{g,y}$$

$$I_z\dot{\omega}_z + (\omega_x + \alpha_x)(\omega_y + \alpha_y)(I_y - I_x) = T_{c,z} + T_{g,z}$$

Finally, we have

$$\dot{\omega}_x = \frac{T_{c,x} + T_{g,x}}{I_x} - \frac{1}{I_x}(\omega_y + \alpha_y)(\omega_z + \alpha_z)(I_z - I_y)$$

$$\dot{\omega}_y = \frac{T_{c,y} + T_{g,y}}{I_y} - \frac{1}{I_y}(\omega_z + \alpha_z)(\omega_x + \alpha_x)(I_x - I_z)$$

$$\dot{\omega}_z = \frac{T_{c,z} + T_{g,z}}{I_z} - \frac{1}{I_z}(\omega_x + \alpha_x)(\omega_y + \alpha_y)(I_y - I_x)$$

We also know the kinematic equation in quaternion representation is given

$$\dot{q} = \frac{1}{2}\Omega(\omega)q \tag{4}$$

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & \omega_z & -\omega_y & \omega_x \\ -\omega_z & 0 & \omega_x & \omega_y \\ \omega_y & -\omega_x & 0 & \omega_z \\ -\omega_x & -\omega_y & -\omega_z & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix}$$

Now consider the state vector $x = \begin{bmatrix} q_1 & q_2 & q_3 & \omega_x & \omega_y & \omega_z \end{bmatrix}^\top$. We can write the attitude dynamics as follows

$$\dot{x} = \begin{bmatrix} \frac{1}{2}(q_2\omega_z - q_3\omega_y q_4\omega_x) \\ \frac{1}{2}(-q_1\omega_z + q_3\omega_x + q_4\omega_y) \\ \frac{1}{2}(q_1\omega_y - q_2\omega_x + q_4\omega_z) \\ \frac{T_x}{I_x} - \frac{1}{I_x}(\omega_y + \alpha_y)(\omega_z + \alpha_z)(I_z - I_y) \\ \frac{T_y}{I_y} - \frac{1}{I_y}(\omega_z + \alpha_z)(\omega_x + \alpha_x)(I_x - I_z) \\ \frac{T_z}{I_z} - \frac{1}{I_z}(\omega_x + \alpha_x)(\omega_y + \alpha_y)(I_y - I_x) \end{bmatrix}$$

Note this is a nonlinear system. Thus, we linearize around the equilibrium point $x = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^\top$

$$\dot{x} = Ax + Bu$$

where

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.5 \\ \frac{6\omega_0^2(I_z - I_y)}{I_x} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{8\omega_0^2(I_z - I_x)}{I_y} & 0 & 0 & 0 & 0 & \frac{\omega_0(I_z - I_x)}{I_y} \\ 0 & 0 & \frac{2\omega_0^2(I_y - I_x)}{I_z} & 0 & 0 & \frac{\omega_0(I_x - I_y)}{I_z} & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{1}{I_x} & 0 & 0 \\ 0 & \frac{1}{I_y} & 0 \\ 0 & 0 & \frac{1}{I_z} \end{bmatrix}$$

The full linearization was computed using SymPy in the following Colab Notebook.

Finally, our product features a gyroscope which provides information about the spacecraft's relative orientation. Therefore, we can deduce our output dynamics to be

$$y = Cx$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

3 System Properties

Stability

We first determine the stability of the uncontrolled system $\dot{x} = Ax$. From MATLAB

```
% Define CubeSat properties
Ix = 0.025;
Iy = 0.025;
Iz = 0.005;

% Define orbital frequency
w0 = 0.0011;

A = [0   0   0   1/2   0   0;
     0   0   0   0   1/2   0;
     0   0   0   0   0   1/2;
     6*w0^2*(Iz-Iy)/Ix   0   0   0   0   0;
     8*w0^2*(Iz-Ix)/Iy   0   0   0   0   w0*(Iz-Ix)/Iy;
     0   2*w0^2*(Iy-Ix)/Iz   0   0   w0*(Ix-Iy)/Iz   0];

% Compute eigenvalues of A
eig(A)
```

```
ans = 6x1 complex
    0.0000 + 0.0000i
    0.0000 + 0.0000i
    0.0000 + 0.0017i
    0.0000 - 0.0017i
    0.0000 + 0.0000i
    0.0000 + 0.0000i
```

Given that $Re(\lambda) = 0$ for all λ, we know the linearization is merely stable. However, we cannot conclude anything about the stability of the nonlinear system.

Controllability

An LTI system is controllable if the controllability matrix $W_c = \begin{bmatrix} B & AB & A^2B & ... & A^{n-1}B \end{bmatrix}$ is full rank. From MATLAB

```
rank(ctrb(A, B))
```

```
ans = 6
```

Thus, we determine the system is controllable.

Observability

An LTI system is observable if the observability matrix $W_o = \begin{bmatrix} C & CA & CA^2 & ... & CA^n - 1 \end{bmatrix}^\top$ is full rank. From MATLAB

```
rank(obsv(A, C))
```

```
ans = 6
```

Thus, we determine the system is observable.

4 Feedback Control

Given the LTI system

$$\dot{x} = Ax + Bu$$
$$y = Cx$$

Consider a controller of the form

$$u = -kx + k_f r$$

where r is the reference signal that we want to drive the system to.
Consider the following quadratic cost function

$$J = \int_0^\infty xQx^\top + uRu^\top dt$$

where Q is the state cost matrix and R is the input cost matrix.

The optimal controller gains that minimize J are given by

$$K = -R^{-1}B^\top P$$

where P is the solution to the following Algebraic Ricatti Equation

$$A\top P + PA - PB^\top R^{-1}BP + Q = 0$$

We compute the LQR using MATLAB's built-in function

```
Q = diag([1 1 1 1 1 1]);
R = diag([10000 10000 10000]);

KLQR = lqr(A, B, Q, R);
```

```
KLQR = 3x6
0.0100 0.0000 -0.0000 0.0187 -0.0000 0.0000
-0.0000 0.0100 0.0000 -0.0000 0.0187 -0.0000
0.0000 -0.0000 0.0100 0.0000 -0.0000 0.0122
```

We discuss the choices of Q and R during the following sections.

Now we find k_f as follows. Write the closed loop system as

$$\dot{x} = Ax + B(-kx + k_f r)$$

$$\dot{x} = (A - Bk)x + Bk_f r$$

We want to drive the system to an equilibrium state x_e such that $\dot{x} = 0$

$$0 = (A - Bk)x_e + Bk_f r$$

$$x_e = -(A - Bk)^{-1} Bk_f r$$

Recall $y = Cx$

$$y = -C(A - Bk)^{-1} Bk_f r$$

Finally, we want to drive $y \to r$

$$r = -C(A - Bk)^{-1} Bk_f r$$

$$1 = -C(A - Bk)^{-1} Bk_f$$

Thus, we have

$$k_f = -(C(A - Bk)^{-1} B)^{-1}$$

From MATLAB

```
Kf = - inv(C/(A-B*KLQR)*B)
```

```
Kf = 3x3
0.0100 0.0000 -0.0000
0.0000 0.0100 0.0000
0.0000 -0.0000 0.0100
```

5 Kalman Filter Observer

Consider an unknown disturbance present in our process and measurement dynamics.

$$\dot{x} = Ax + Bu + w$$

$$y = Cx + v$$

Consider w and v to be drawn from a Gaussian distribution of zeros mean with covariances Rw and Rv, respectively.

We seek to design an observer of the form

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$$

The optimal observer gain L that minimize the prediction mean squared error $E\left[\frac{1}{2}||x - \hat{x}||^2\right]$ is given by

$$L = PC^\top R_w^{-1}$$

where

$$\dot{P} = AP + PA6\top - PC^\top R_w^{-1} CP + R_V$$

$$P = E[(x - \hat{x}(x - \hat{x}^\top]$$

From MATLAB

```
Rw = 0.001;
Rv = 0.0001;

L = lqr(A', C', Rw, Rv)';
```

```
L = 6x3
3.6280 -0.0000 -0.0000
-0.0000 3.6280 -0.0004
-0.0000 -0.0004 3.6280
3.1623 0.0000 0.0000
-0.0000 3.1623 -0.0036
-0.0000 -0.0028 3.1623
```

6 Closed Loop System

Using SIMULINK, we simulate the entire closed loop system to assess the performance of our controller and observer design.

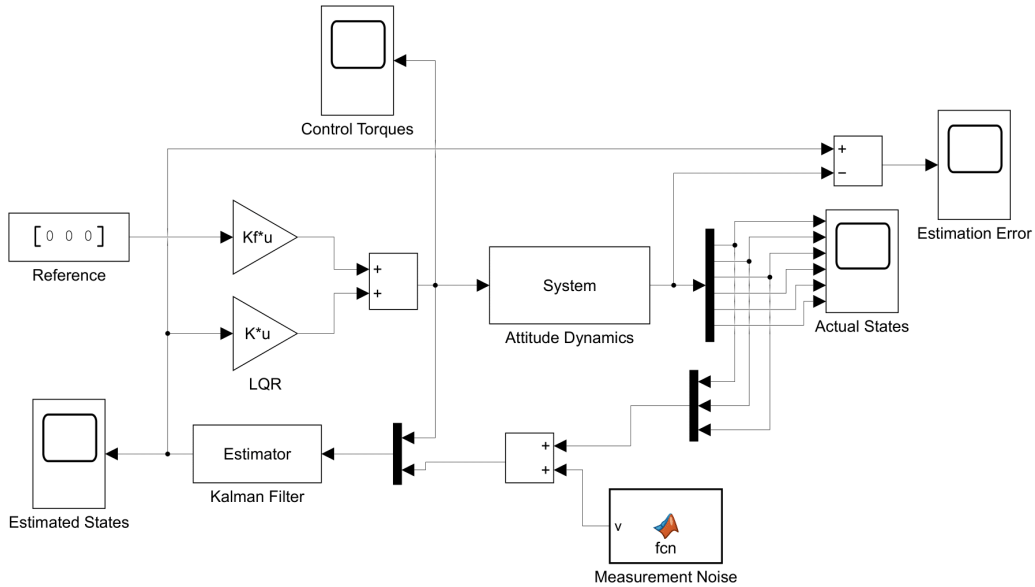

Figure 1: SIMULINK Model of Closed Loop System

The plant and observer blocks were models as LTI systems using the MATLAB State Space Model.

```
B = [0    0    0;
0    0    0;
0    0    0;
1/Ix    0    0;
0    1/Iy    0;
0    0    1/Iz];

C_full = eye(6);

D = 0;

% Define plant as ss model
System = ss(A, B, C_full, D);
```

```
Cw = 0.001;
Cv = 0.0001;

L = lqr(A', C', Cw, Cv)';

% Augment system to take u and y as inputs
A_k = A - L*C;
B_k = [B L];
C_k = C;
D_k = 0;

% Define observer as ss model
Estimator = ss(A_k, B_k, C_k, D_k);
```

Below is the system response to an initial attitude $\varphi = 15 \quad \theta = -15 \quad \psi = 30$ and angular velocity $\omega_x = 0.005\ rad/s \quad \omega_y = 0\ rad/s \quad \omega_z = 0.001\ rad/s$

In quaternion representation this translates to $x_0 = \begin{bmatrix} -0.6634 & -0.5903 & 0.4584 & 0.005 & 0 & 0.001 \end{bmatrix}$

Consider a measurement noise covariance of $R_w = 0.00001$

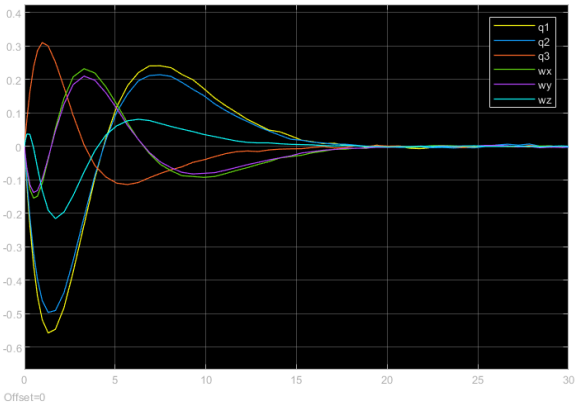

Figure 2: Estimated States

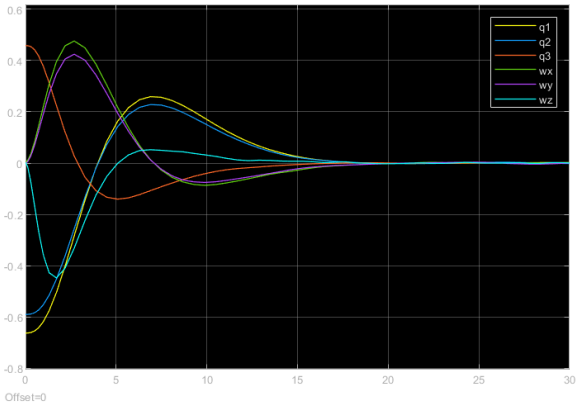

Figure 3: Actual States

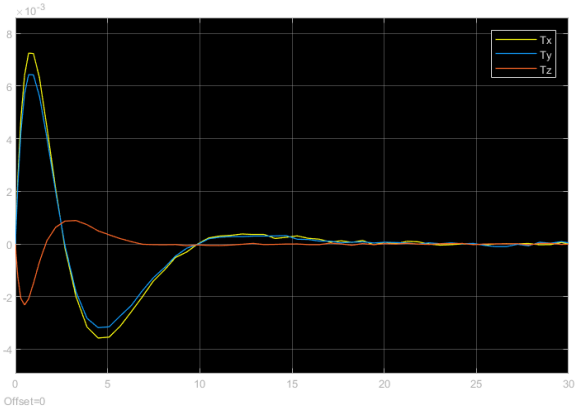

Figure 4: Reaction Wheel Control Torques

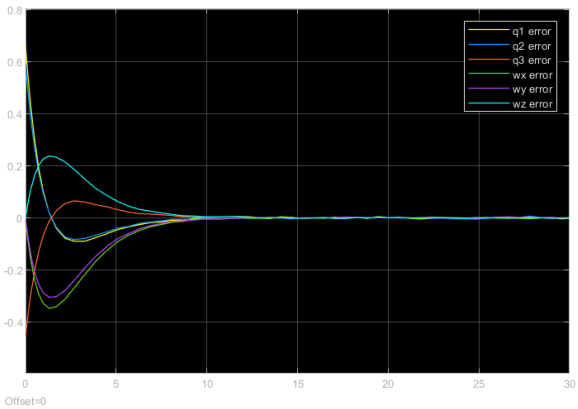

Figure 5: Estimation Error

We observe the system is able to drive the CubeSat to an equilibrium state at the origin, stabilizing its angular velocity and the desired pointing reference. We also observe the Kalman Filter observer is able to convergence to the actual system states.

Furthermore, we observe the control torques are within a reasonable scale 10 mNm, well within the capability of most reaction wheels. This numbered was modeled during the selection of the state and input costs used to compute the LQR controller gains. Looking back, we required an input cost 1000 greater than the state cost to achieve this control behavior.

Now we demonstrate the robustness of our system to higher measurement disturbances. Consider a measurement noise covariance of $R_w = 0.001$

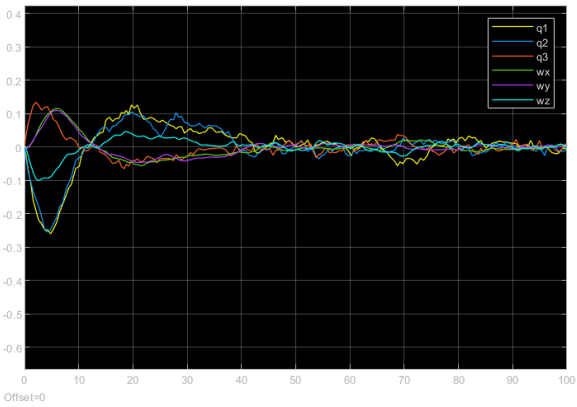

Figure 6: Estimated States

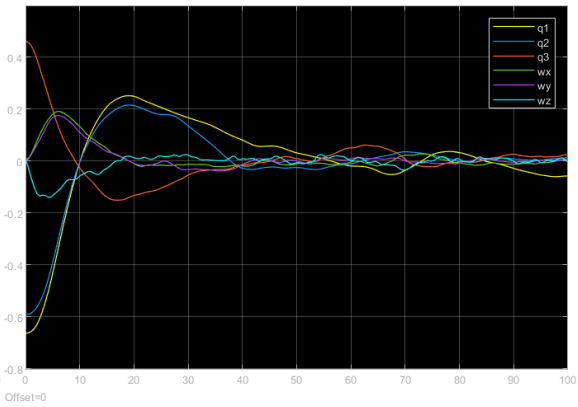

Figure 7: Actual States

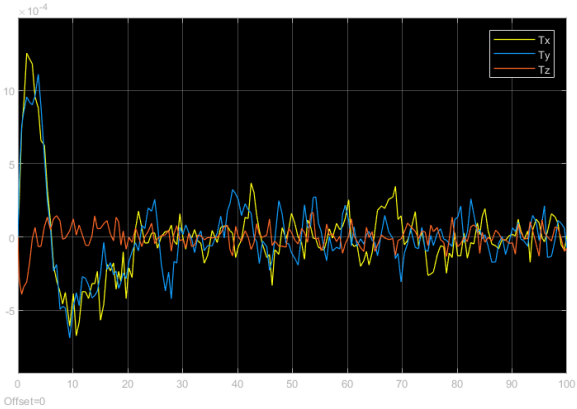

Figure 8: Reaction Wheel Control Torques

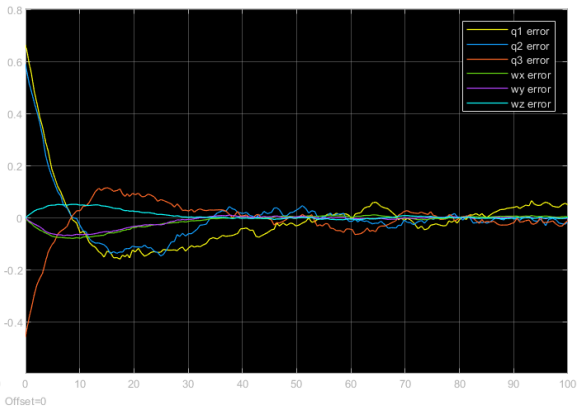

Figure 9: Estimation Error

Despite the higher noise levels, we observe our controller is still able to drive the system to the origin, on a longer time window. Similarly, we the observer estimation takes longer to converge to the true states.

Finally, we demonstrate the ability of our ADCS system to drive the CubeSat to a desired pointing reference. Starting at the origin, the following demonstrates controller acting towards the desired reference $\varphi = 15 \quad \theta = -15 \quad \psi = 30$, which in quaternion representation becomes $r = \begin{bmatrix} -0.6634 & -0.5903 & 0.4584 \end{bmatrix}$

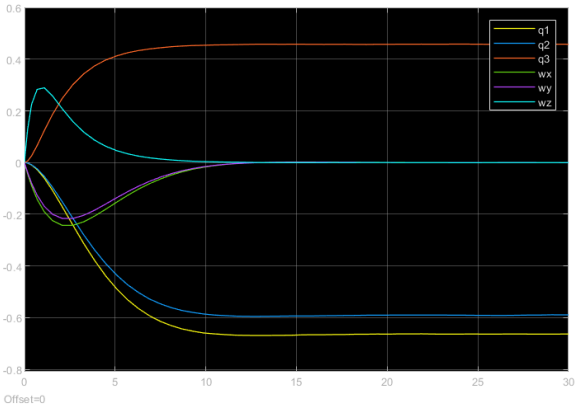

Figure 10: Estimated States

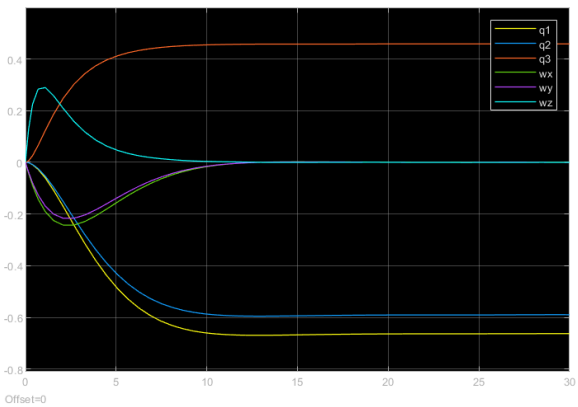

Figure 11: Actual States

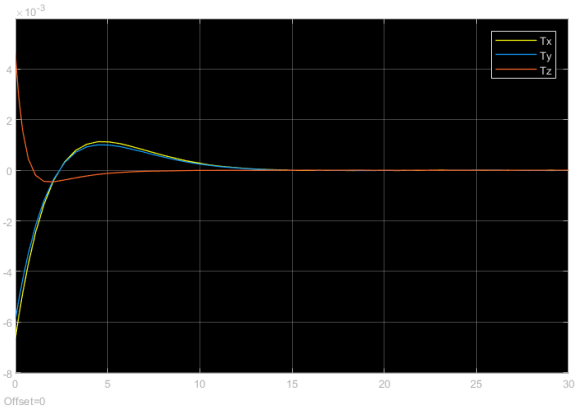

Figure 12: Reaction Wheel Control Torques

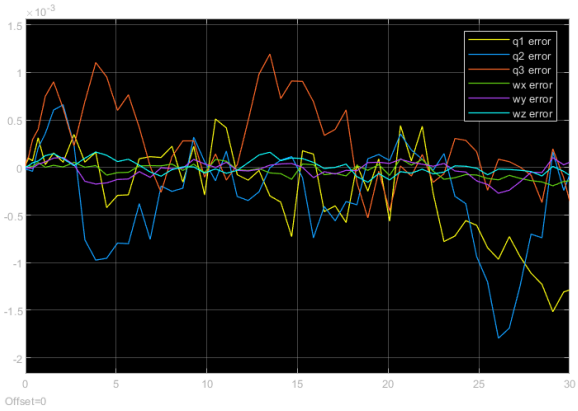

Figure 13: Estimation Error

As expected, we observe our controller is able to drive the spacecraft from the origin to the desired pointing attitude. Furthermore, given the initial condition of our spacecraft matched our initial estimation guess (the origin), we observe our estimation error is bounded at 10^-3, introduced by the measurement noise.